

ELIO KOKA

HARDWARE & SOFTWARE INTEGRATION ENGINEER

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Senior Systems Integration Engineer with 8+ years of experience in system-level debugging, automated testing, and cross-functional root cause analysis. Specializes in scaling complex humanoid and mobile robotic architectures from prototype to production. Proven track record of leading the bring-up of advanced compute, perception, and electro-mechanical sub-systems while building scalable hardware-in-the-loop (HIL) test architectures.

PROFESSIONAL EXPERIENCE

AUTOMATION HARDWARE & SOFTWARE INTEGRATION ENGINEER

Foundation | Sep 2025 - Present

- Architected a full-stack Manufacturing Execution System (MES) from scratch to scale humanoid production, integrating live product genealogy, BOM/inventory management, and interactive 3D work instructions.
- Automated hardware integration workflows within the MES to execute actuator calibration, subassembly validation, and EOL testing, auto-logging QC metrics for rapid multi-system root cause analysis.
- Led the hardware/software bring-up of next-generation humanoids, spearheading the core compute migration to NVIDIA Thor and engineering a parallel CAN-based flashing system that cut firmware upgrades from 1.5 hours to <3 minutes.

ROBOTICS TEST ENGINEER

Wheel.me | Apr 2024 - May 2025

- Architected Python and Azure automated diagnostic pipelines to systematically triage autonomous guided vehicle (AGV) fleet anomalies, reducing system downtime by 20% and manual validation time by 33%.
- Executed FMEA and nominal path functional tests for safety-critical mobility systems, actively diagnosing CAN protocol failures to ensure system compliance and stability.
- Verified hardware/software interfaces during early development, tracking system-level performance and stability to provide continuous data-driven feedback for design iterations.

ROBOT BRING-UP ENGINEER

Carbon Robotics | Jun 2022 - Apr 2024

- Managed system bring-up and end-of-line validation for complex AI-powered autonomous agricultural robots, ensuring robust electro-mechanical functionality across mobility, perception, and power subsystems.
- Revolutionized manufacturing test workflows by automating integration tasks with Python and Bash, driving an 83% reduction in robotic bring-up times.
- Designed and implemented custom diagnostic tools and instrumentation, including a Raspberry Pi testing jig to characterize sensors and servos, enhancing verification processes for high-performance platforms.

IT SYSTEMS SUPPORT ENGINEER & TECHNICIAN

TechSmith Corporation & Michigan State University | Aug 2017 - Aug 2021

- Automated OS deployment pipelines and managed complex network infrastructure (RAID arrays, external compute units), driving rapid triage of cross-functional system issues across enterprise environments.
- Coordinated closely with software development and QA teams to triage user bugs, deploy technical patches, and document systematic workarounds in internal knowledge bases.

CORE SKILLS & TOOLS

- **Systems Engineering:** System Bring-up & Integration, Hardware/Software Interfaces, Root Cause Analysis, FMEA, EOL Testing, Mechatronics
 - **Sub-Systems:** Onboard Compute (ARM, x86, RISC-V), Perception, Sensors, Servos/Actuation, BMS, CAN
 - **Tools & Software:** Python, C, MongoDB, S3, Bash, Git, CircleCI, Azure, Data Logging, CI/CD, Automated Testing
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EDUCATION

MICHIGAN STATE UNIVERSITY

Bachelor of Science, Computational Mathematics | Focus in Machine Learning - 2017 - 2022